

# Attrition Prevention Suggestions

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## Purpose

These suggestions come from the attrition analysis of 1,470 employees (16.1% attrition) using statistical tests, a predictive model and SHAP explanations. Each action is linked to evidence from the data.

## Recommended actions

| # | Risk factor (evidence)                                                                                         | Recommended action                                                                                                              | Measure of success                                                  |
|---|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1 | <b>Overtime:</b> 30.5% attrition with overtime vs 10.4% without; top SHAP driver                               | Audit workload in high-overtime roles (Sales, Laboratory Technician); cap monthly overtime; add temporary staff in peak periods | Overtime attrition falls toward 20% within 12 months                |
| 2 | <b>Low pay:</b> lowest income band 29.3% vs about 10% in higher bands; low stock options raise risk            | Review pay for the lowest band and people below their job-level average; extend stock-option or bonus eligibility               | Fewer exits in lowest income band; pay gap to level average reduced |
| 3 | <b>Early tenure:</b> 29.8% attrition in the first 2 years; young age and many past employers add risk          | Structured onboarding, buddy or mentor, 30/90/180-day check-ins                                                                 | First-2-year attrition below 20%                                    |
| 4 | <b>High-risk roles:</b> Sales Representatives 39.8%                                                            | Role-specific review: targets, commission structure, career path to Sales Executive                                             | Role attrition below 30%                                            |
| 5 | <b>Travel and commute:</b> frequent travel and long distance from home raise risk                              | Offer remote or hybrid days, travel rotation, commute or relocation support                                                     | Lower risk scores in travel-heavy roles                             |
| 6 | <b>Low satisfaction:</b> low job, environment and work-life satisfaction increase risk                         | Run quarterly pulse surveys; act on team-level results; manager training                                                        | Satisfaction scores rise; fewer low scorers                         |
| 7 | <b>Known high-risk people:</b> model flags 72% of leavers (High bucket: 37.7% actual attrition vs 4.2% in Low) | HR holds monthly stay-interviews with the High-risk list; re-score employees monthly                                            | Share of High-risk employees who resign falls                       |

## Expected business impact

Assuming replacement cost is 50% of annual salary, the 237 employees who left cost about \$6.8M. The model placed 171 of them (72%) in the High-risk bucket, worth about \$4.1M. If targeted retention saves 30% of such people, the estimated saving is about **\$1.2M per year**. Both the 50% cost and the 30% success rate are assumptions and can be changed in the notebook.

## Implementation roadmap

| Timeline       | Actions                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------|
| First 30 days  | Share the High-risk list with HR; start stay-interviews; audit overtime hours                                 |
| 30 to 90 days  | Pay review for lowest band; launch onboarding programme; pilot flexible work for travel-heavy roles           |
| 90 to 180 days | Pulse surveys; manager training; re-train the model with new data and track results on the Power BI dashboard |

## Notes and limitations

The dataset is a fictional practice dataset from IBM. Findings show association, not proof of cause, so pilot each action and measure results. Risk scores should support, not replace, manager judgement, and should be used to offer help (not to penalise employees).